

Task 4 Report – Dockerize and Deploy Product API

Title

Task 4 – Dockerize and Deploy Product API using Docker

Objective

The objective of this task is to containerize the Spring Boot Product API application using Docker and Docker Compose. This enables the application to run consistently across different environments without manual dependency setup.

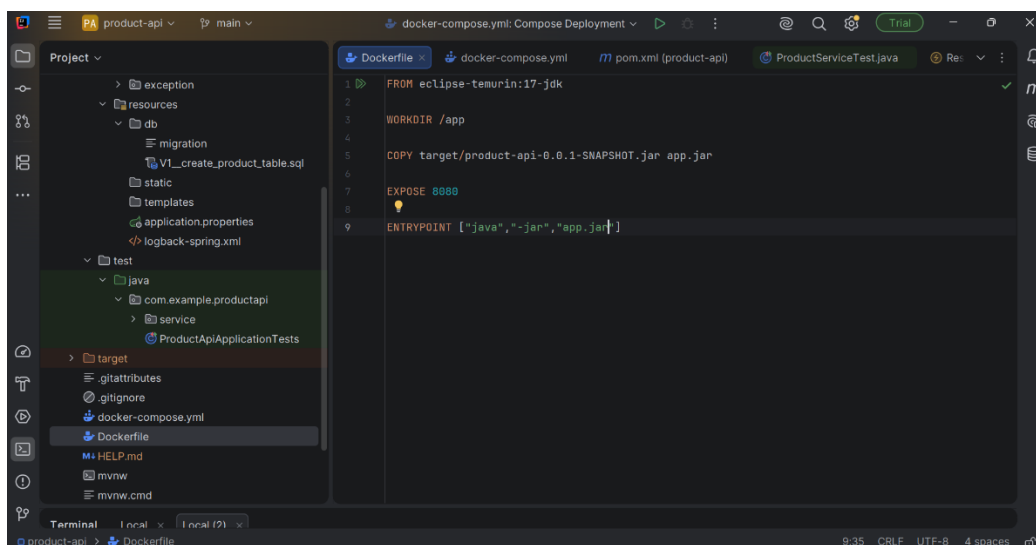
Technologies Used

- Java 17
- Spring Boot 3.5.3
- Spring Data JPA
- MySQL
- Docker
- Docker Compose
- Maven
- Git & GitHub

Implementation

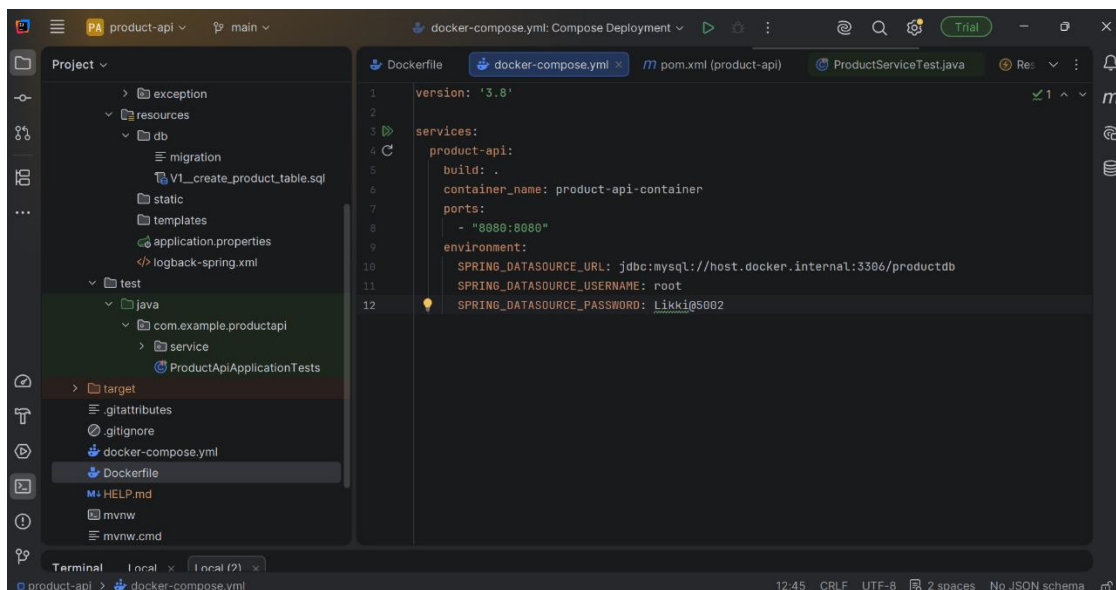
Step 1 – Dockerfile Creation

Created a Dockerfile to package the Spring Boot application into a Docker image.



Step 2 – Docker Compose Configuration

Created a docker-compose.yml file to configure and run the application.

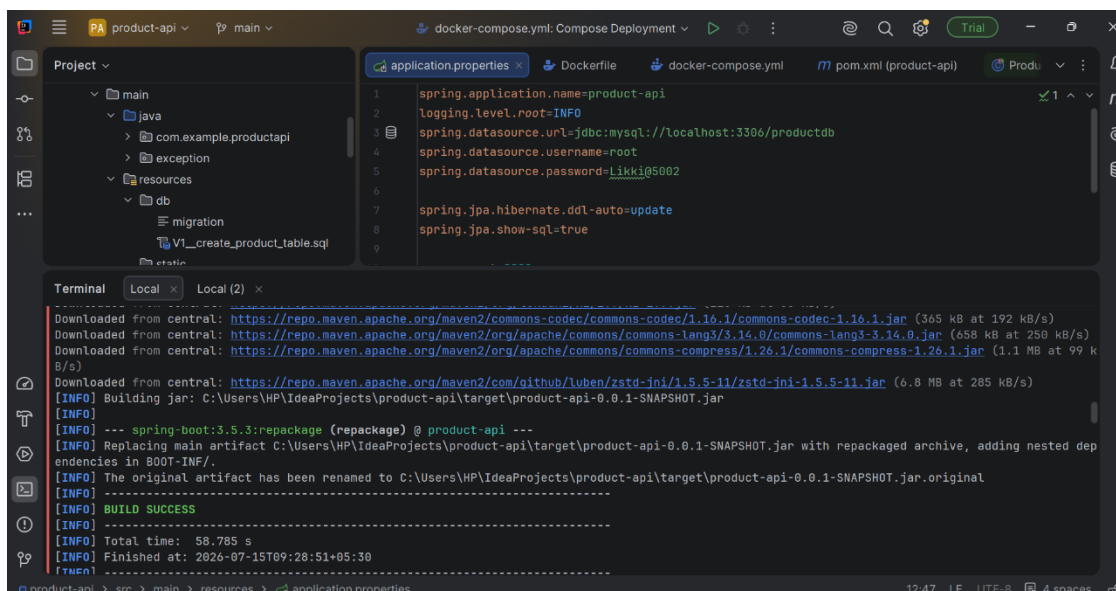


Step 3 – Build the Project

Generated the executable JAR file using Maven.

Command:

.\mvnw.cmd clean package



Step 4 – Build Docker Image

Built the Docker image and created the application container.

Command: docker compose up –build


```
"name":"Gaming Laptop",  
"description":"HP Victus",  
"price":65000.0,  
"quantity":5  
}  
]
```

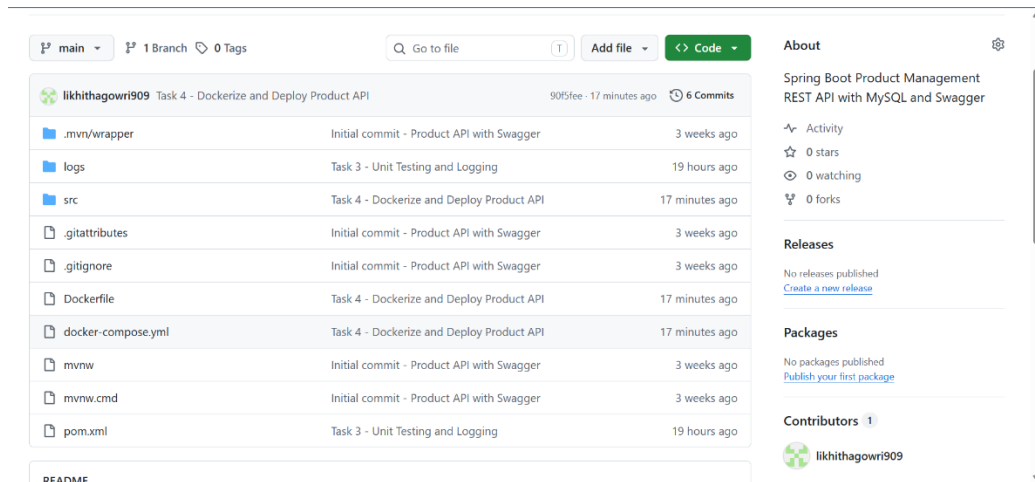


Step 7 – GitHub Repository

Uploaded the completed Dockerized project to GitHub.

Repository includes:

- Dockerfile
- docker-compose.yml
- Spring Boot source code
- Flyway migration
- Configuration files



Outcome

- Successfully containerized the Spring Boot Product API.
- Built the Docker image successfully.
- Executed the application inside a Docker container.
- Connected the application with the MySQL database.
- Successfully tested the REST API.
- Uploaded the completed project to GitHub.

Conclusion

Task 4 was successfully completed by Dockerizing the Spring Boot Product API application and deploying it using Docker. The application was successfully built, executed inside a Docker container, connected to the MySQL database, and verified through REST API testing. The complete project has been documented and uploaded to GitHub.